

$$10x^2 + 2y^2 + 8xy - 92x - 40y + 218 = 0$$

$$[x \ y] \begin{bmatrix} 10 & 4 \\ 4 & 2 \end{bmatrix} + 2 \begin{bmatrix} -46 & -20 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + 218 = 0$$

$$[x \ y] \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix} + 2 \begin{bmatrix} -23 & -10 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + 109 = 0$$

$$\Rightarrow \begin{vmatrix} 5 & 2 & -23 \\ 2 & 1 & -10 \\ -23 & -10 & 109 \end{vmatrix} \Rightarrow \begin{vmatrix} 5 & 2 & -23 \\ 2 & 1 & -10 \\ -3 & 0 & 9 \end{vmatrix}$$

$$R_3 = R_3 + 10R_2$$

$$\begin{array}{r} 23 \\ 5 \\ \hline 105 \end{array}$$

$$\Rightarrow \begin{vmatrix} 5 & 2 & -23 \\ 2 & 1 & -10 \\ -3 & 0 & 9 \end{vmatrix} \Rightarrow \begin{vmatrix} 5 & 2 & -8 \\ 2 & 1 & -4 \\ -3 & 0 & 9 \end{vmatrix}$$

$$C_3 = C_3 + 3C_1$$

$$\Rightarrow \begin{vmatrix} 5 & 2 & -8 \\ 2 & 1 & -4 \\ -3 & 0 & 0 \end{vmatrix} \Rightarrow \begin{vmatrix} 5 & 2 & 0 \\ 2 & 1 & 0 \\ -3 & 0 & 0 \end{vmatrix}$$

$$C_3 = C_3 + 4C_2$$

∴ determinant is 0.

$\Rightarrow$ Singular matrix:-
represents pair of straight lines:-

$$x + y = 7$$

$$y + 3x = 13$$

$$\Rightarrow \text{Coeff: } \begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix}$$

$$\downarrow R_3 = R_3 - 3R_2$$

$$\text{Augmented: } \left[\begin{array}{cc|c} 1 & 1 & 7 \\ 3 & 1 & 13 \end{array} \right]$$

$$\left[\begin{array}{cc|c} 1 & 1 & 7 \\ 0 & -2 & -8 \end{array} \right] \text{Rank } 2$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 7 & \\ 0 & -2 & -8 & \end{array} \right] \text{Rank } = 2$$

unique sol:

$$-2y = -8 \Rightarrow y = 4$$

$$x + y = 7 \Rightarrow x = 3$$

$$\text{Solution: } \vec{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$